

Problem 11.1

$$J = P_{21} + P_{31} + P_{32} - P_{32} P_{21} - P_{21} P_{32}$$

$$C_1 P_1 + \dots + C_5 P_5 = 0 \Rightarrow C_1 = C_2 = C_3 = 0 \wedge C_4 = C_5 = 0$$

Problem 11.2

$$a) X = \begin{bmatrix} a & b & c \\ a & b & c \\ a & b & c \end{bmatrix}$$

$$b) \begin{bmatrix} a & b & c \\ d & e & f \\ -a-d & -b-e & -c-f \end{bmatrix}$$

$$c) N=3$$

$$C=6$$

$$N+C=9$$